

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	18 june 2026
Team ID	
Project Name	SHOPEZ : E-commerce Application
Maximum Marks	4 Marks

Technical Architecture:

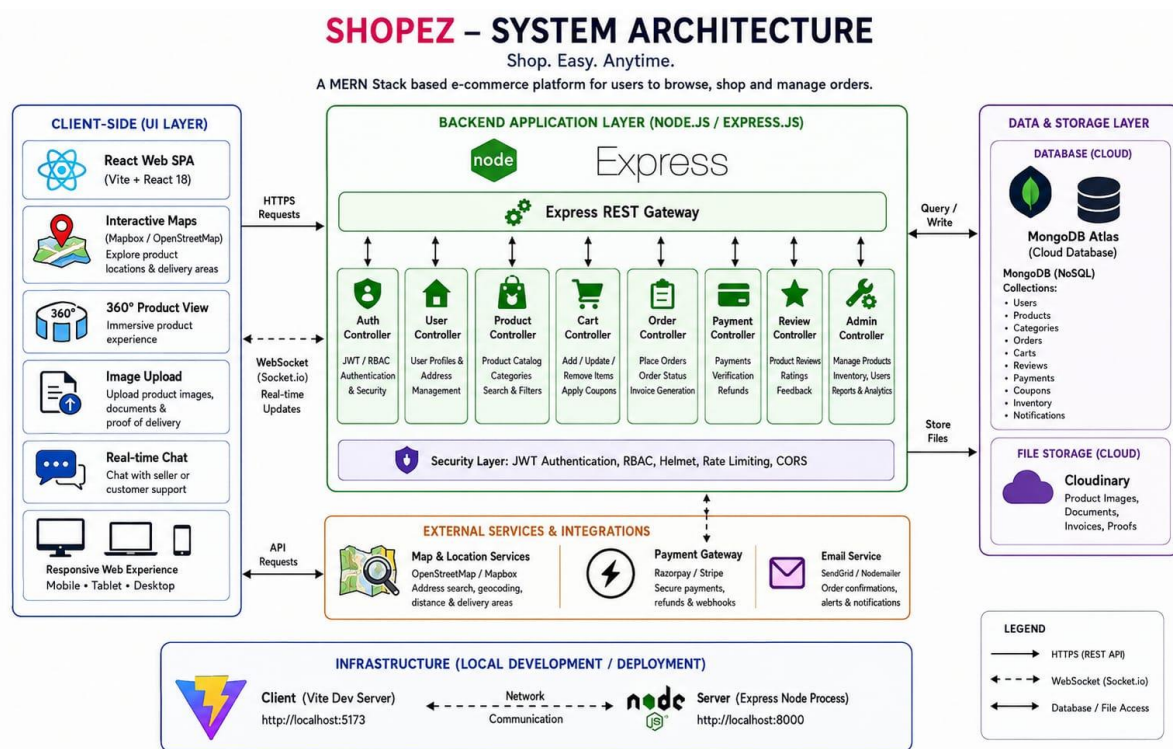


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Responsive Web UI providing the shop storefront, product details, cart modifications, checkouts, and admin database controllers.	HTML5, CSS3, JavaScript, React.js (v19), Vite, Lucide React (Icons)
2	Application Logic-1	Core backend server routing REST API endpoints for user accounts, products cataloging, cart CRUD, checkouts, and admin operations.	Node.js, Express.js (v5)
3	Application Logic-2	Encryption and Authorization logic to securely hash user passwords and sign/verify session authentication tokens.	bcryptjs, jsonwebtoken (JWT)
4	Application Logic-3	In-memory query logic for sorting catalogs, calculating bill structures (GST + convenience fees), and catalog auto-seeding.	Vanilla JavaScript ES6
5	Database	NoSQL Database storing documents for Users, Products, Carts, Orders, and Admins.	MongoDB, Mongoose ODM
6	Cloud Database	Cloud-based managed NoSQL database cluster storing production collections.	MongoDB Atlas
7	File Storage	Local directory storage for project log files, product assets, and cached build documents.	Local Filesystem / Node.js path & fs packages
8	External API-1	Image resource service providing high-definition placeholder images for catalog products and banners.	Unsplash Image Source API
9	External API-2	Payment simulation checks for simulated UPI, Cards, and Cash on Delivery order checkouts.	Mock Payment Gateway API
10	Machine Learning Model	Product recommendation engine that checks item attributes to suggest trending or high-discount products.	Not Applicable (Custom mathematical logic filtering is used)

S.No	Component	Description	Technology
11	Infrastructure (Server / Cloud)	Local Configuration: Node.js server (port 8000), Vite dev server (port 5173), MongoDB service (port 27017). Cloud Configuration: Vercel (Frontend), Render (Backend), MongoDB Atlas (Database).	Local Server, MongoDB Daemon, Vercel CDN, Render Hosting

Table-2: Application Characteristics:’

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	The application is built using standard, widely-adopted open-source frameworks to ensure stability, quick development, and modular components.	React.js (v19), Express.js (v5), Mongoose ODM (v9)
2	Security Implementations	• Password Hashing: Customer and Admin passwords are encrypted before database storage. • Session Authentication: Token-based validation checks on protected paths. • Access Guards: Dedicated authorization middleware restricts administrative endpoints.	bcryptjs (Hashing), jsonwebtoken (JWT Bearer tokens), Express Route Guards (protect, isAdmin)
3	Scalable Architecture	Decoupled 3-Tier architecture separating: 1. Presentation Layer (Vite + React UI client) 2. Application Tier (Express REST API) 3. Data Tier (MongoDB Database) This permits independent tier scaling.	React Web Bundles, Node.js Server Environment, MongoDB Document Store
4	Availability	• Stateless Express server design allows horizontal deployment across multiple server instances behind load balancers. • Client UI remains online and handles API connection drops gracefully using fallback error banners.	PM2 Clustering, Docker Staging, Nginx Load Balancers, MongoDB Replica Sets

S.No	Characteristics	Description	Technology
5	Performance	• Lightweight Vite bundler for frontend assets. • Mongoose database queries utilize indices for quick product searches. • Backend controllers use parallel asynchronous processing (Promise.all) for high throughput.	Vite compilation, MongoDB Database Indexing, Vercel CDN static caching